A Controlled Cross-Dataset Evaluation of Lightweight YOLO Generations for Breast Ultrasound Lesion Segmentation

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Abstract. Breast ultrasound (BUS) lesion segmentation models can perform well on internal data yet degrade when scanner characteristics and acquisition conditions change. This study compares the cross-dataset behavior of three nano-scale YOLO segmentation generations, YOLOv8n-seg, YOLO11n-seg, and YOLO26n-seg, under a common-source protocol. All models were developed only on BUS-BRA and then frozen for evaluation on the BUS-BRA test set and the independent BUSI and BUS-UCLM datasets. Performance was assessed using Ultralytics mask precision, recall, and mean average precision (mAP), together with image-level Dice and Intersection over Union (IoU). Uncertainty was estimated with 10,000-resample bootstrap 95% confidence intervals, while paired Dice differences were tested using two-sided Wilcoxon signed-rank tests with Holm correction. BUS-BRA Dice was consistently high (0.9010-0.9111), but all models degraded externally. YOLO11 achieved the highest average external Dice (0.6560) and the smallest average generalization gap (0.2542), while YOLO26 obtained the highest in-domain Dice and the lowest model complexity. On BUSI, YOLO11 and YOLO26 significantly outperformed YOLOv8 after Holm correction, whereas YOLO11 and YOLO26 did not differ significantly. These results show that small source-domain advantages do not reliably predict cross-dataset behavior and support routine independent external validation of BUS segmentation models.

Keywords: breast ultrasound · lesion segmentation · YOLO · cross-dataset generalization · domain shift · external validation

1 Introduction

Breast cancer remains a major global health burden, with approximately 2.4 million new diagnoses among women and about 694,000 deaths reported worldwide in 2024 [24]. Ultrasound is widely used as a complementary breast-imaging modality because it is non-ionizing, relatively inexpensive, real-time, and useful in dense breast tissue. Automated BUS analysis, however, remains difficult: lesion appearance varies in shape, size, texture, and echogenicity, while speckle

noise, low contrast, acoustic artifacts, and ambiguous boundaries complicate delineation [25, 28]. Reliable lesion segmentation is therefore important not only for localization, but also for quantitative lesion analysis and computer-aided decision support.

U-Net and related encoder–decoder models established a strong baseline for biomedical image segmentation [15]. More recent BUS methods have focused on stronger feature interaction and robustness to lesion variability, including cross-image dependency modeling and invariant feature learning [11, 25]. Despite these advances, reported performance remains sensitive to dataset composition, scanner characteristics, annotations, partitioning, and metric choice [17, 28]. This issue becomes important at deployment, where the image distribution may differ from the data used for development. Scanner-agnostic segmentation and external-validation studies have accordingly emphasized that strong internal accuracy alone is not sufficient evidence of robustness [10, 26].

YOLO provides an efficient instance-segmentation framework and has already been applied to breast ultrasound tumor segmentation [12]. Recent Ultralytics releases provide segmentation variants for YOLOv8, YOLO11, and YOLO26 [7–9]. The closest prior benchmark by Yu et al. compared twelve YOLO variants on BUSI and BUS-UCLM and reported substantial Dice degradation under cross-database transfer [27]. Their design used BUSI and BUS-UCLM as both source and target domains in cross-training. A related question remains: when several lightweight YOLO generations are developed from the same source dataset, do their relative advantages persist on multiple unseen datasets?

We used BUS-BRA as the only development dataset and evaluated the three frozen models on the BUS-BRA test set, BUSI, and BUS-UCLM [1, 3, 19]. The comparison combines instance-level mAP with image-level Dice/IoU, bootstrap confidence intervals, paired Wilcoxon tests with Holm correction, and a Dice-based generalization gap. Unlike prior cross-database studies, our protocol fixes BUS-BRA as the common development source and evaluates the frozen models on independent BUS datasets without target-domain adaptation, allowing us to examine whether their relative performance is preserved under distribution shift.

The remainder of this paper is organized as follows. Section 2 reviews related work on BUS segmentation and cross-dataset evaluation. Section 3 describes the datasets, experimental settings, evaluation protocol, and statistical analysis. Section 4 presents the experimental results, followed by a discussion of the findings in Section 5. Section 6 concludes the paper.

2 Related Work

BUS segmentation has gradually shifted from conventional image-processing methods to deep encoder–decoder networks and, more recently, attention-based architectures. U-Net established a widely used encoder–decoder design for biomedical image segmentation [15], and many later BUS methods have extended this idea to better handle the characteristics of ultrasound images. In particular, lesion boundaries in BUS images are often unclear because of speckle noise, low

contrast, heterogeneous echogenicity, and large variations in lesion shape and size. Recent studies have therefore focused on improving feature interaction and representation robustness. BUSSeg incorporated within-image and cross-image dependency modeling to capture relationships among lesion appearances [25], while Manh et al. combined global adversarial learning with a region-invariant loss to encourage more stable lesion representations [11]. These studies illustrate a broader shift from simply improving segmentation accuracy toward learning features that remain reliable under difficult ultrasound conditions.

Attention-based designs have also become increasingly common in BUS segmentation. TaiChiNet introduced negative-positive cross-attention to strengthen the distinction between lesion and surrounding tissue [20]. DMAeEDNet explored an attention-enhanced encoder–decoder architecture with an emphasis on both segmentation accuracy and computational efficiency [16]. Other recent methods have investigated cross-scale information and boundary-sensitive representations. For example, a cross-scale attention-based U-Net was proposed to improve the integration of local and contextual lesion information [21], while EMGANet emphasized edge-aware and multi-scale attention for breast ultrasound segmentation [6]. Although these methods differ in architecture, they share a common objective: reducing segmentation errors caused by weak boundaries, variable lesion appearance, and ultrasound-specific artifacts.

Alongside model development, benchmark studies have shown that segmentation performance is strongly influenced by the way datasets are constructed and evaluated. BUSIS was introduced to support more consistent comparison of BUS segmentation methods [28]. BUS-Set later combined public BUS images acquired from multiple scanner types and evaluated several deep segmentation architectures under a reproducible protocol [17]. These benchmark efforts showed that reported performance can change with dataset composition, scanner characteristics, partitioning strategy, annotation quality, and metric selection. As a result, high performance on one internal split should not automatically be interpreted as evidence that a model will perform equally well on independently acquired BUS data. This motivates evaluation settings in which development data and external test data are kept clearly separated.

YOLO-based BUS segmentation has emerged more recently. Recent Ultralytics generations extend the YOLO framework beyond object detection to instance segmentation, including YOLOv8, YOLO11, and YOLO26 [7–9]. Mostafa et al. demonstrated that YOLOv8 can be applied effectively to breast ultrasound tumor segmentation on BUSI [12]. A more directly related study was reported by Yu et al., who compared multiple YOLO variants on BUSI and BUS-UCLM and examined both same-database and cross-database performance [27]. Their results showed that models with strong performance within the development dataset could experience substantial Dice degradation when transferred to another BUS dataset. This finding is particularly relevant to lightweight YOLO models, because small differences observed on a source dataset may not preserve the same ranking after domain transfer.

Cross-dataset robustness has also been studied using methods that explicitly address domain shift. DAUS-Net targeted scanner-agnostic domain generalization and investigated whether ultrasound segmentation models could retain performance across scanners [10]. Wang et al. addressed distribution changes at deployment through prototype bank-driven test-time adaptation [22]. Retrieval-based SAM2 has also been explored as a training-free approach for BUS segmentation [4]. These methods are important because they attempt to improve segmentation after encountering a new data distribution. The objective of the present work is different. No target-domain adaptation, fine-tuning, or confidence-threshold optimization is performed. Instead, the selected checkpoints are frozen before external evaluation. This provides a direct view of how much of the source-domain performance is preserved when the same model is exposed to unseen BUS distributions.

Despite the growing number of BUS segmentation methods, controlled comparisons of successive lightweight YOLO generations under a common-source, multiple-target setting remain limited. Previous work has largely focused on improving within-dataset accuracy, comparing heterogeneous segmentation architectures, or adapting models to a target domain. In addition, reporting only aggregate performance measures does not indicate how uncertain the estimated differences are or whether one model consistently performs better than another on the same cases [14]. External validation is therefore more informative when average performance is considered together with uncertainty estimates and paired statistical comparisons [26]. In this study, YOLOv8n-seg, YOLO11n-seg, and YOLO26n-seg are developed using the same BUS-BRA source dataset and evaluated with frozen checkpoints on BUS-BRA test, BUSI, and BUS-UCLM. The analysis combines instance-level mAP, image-level Dice and IoU, bootstrap 95% confidence intervals, paired Wilcoxon signed-rank tests with Holm correction, and a Dice-based generalization gap. This design allows the comparison to focus not only on which model performs best internally, but also on whether its relative advantage is preserved when the data distribution changes.

3 Methodology

The study followed a source-development/external-validation design. BUS-BRA was used exclusively for model development, whereas BUSI and BUS-UCLM served as independent external test sets. Neither external dataset was used for training, checkpoint selection, hyperparameter tuning, confidence-threshold tuning, or fine-tuning. Figs. 1 and 2 illustrate the model-development and frozen-checkpoint evaluation stages, respectively.

3.1 Datasets

Three public breast ultrasound (BUS) datasets were used in this study: BUS-BRA for model development and in-domain evaluation, and BUSI and BUS-UCLM for external evaluation. We split BUS-BRA case-wise into training, validation, and



Fig. 1. Source-domain model development using BUS-BRA for training and validation-based checkpoint selection.

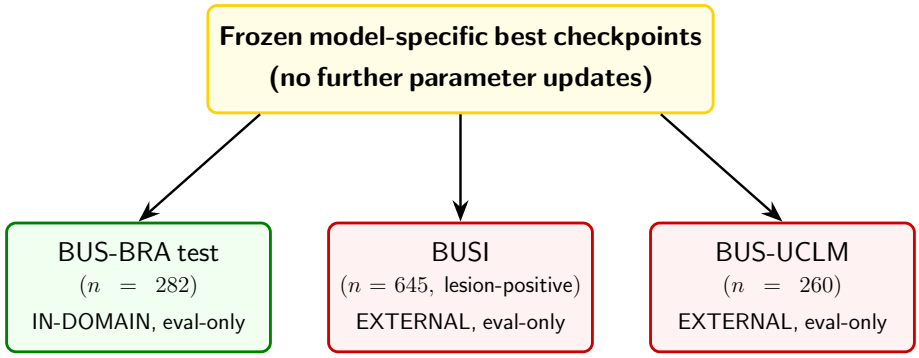


Fig. 2. Frozen-checkpoint evaluation on the BUS-BRA test set and the external BUSI and BUS-UCLM datasets. The yellow box denotes the frozen model-specific checkpoints, the green box denotes in-domain evaluation, and the red boxes denote evaluation on external unseen-domain datasets.

test sets, while using the external datasets without further model tuning. All lesion-positive images were treated as a single lesion class regardless of pathology. Table 1 summarizes the datasets and their experimental roles.

- **BUS-BRA:** BUS-BRA contains 1,875 BUS images from 1,064 patients with lesion masks and clinical metadata [3]. A case-aware split prevented images from the same case from crossing development partitions, yielding 1,323 training, 270 validation, and 282 test images. Benign/malignant counts were 895/428, 179/91, and 194/88, respectively. Because the task is lesion segmentation rather than pathology classification, we mapped all lesions to a single segmentation class.
- **BUSI:** BUSI contains 780 images: 437 benign, 210 malignant, and 133 normal [1]. We excluded normal images because the external benchmark targets lesion-positive segmentation. We audited the release for duplicate images, missing or multiple masks, and image-mask consistency. Following a previously reported concern about BUSI duplicates [13], we removed one byte-identical pair with conflicting pathology labels and inconsistent masks. Multiple masks belonging to the same image were unioned before polygon extraction to preserve all annotated lesion regions. The final external set contained 645 images and 660 lesion instances.

Table 1. Summary of datasets and experimental subsets.

Dataset/subset	Images	B	M	N	Use
BUS-BRA Train	1,323	895	428	0	Training
BUS-BRA Val.	270	179	91	0	Selection
BUS-BRA Test	282	194	88	0	In-domain
BUSI Original	780	437	210	133	–
BUSI Final	645	–	–	0	External
BUS-UCLM Original	683	174	90	419	–
BUS-UCLM Final	260	–	–	0	External

B: benign; M: malignant; N: normal. BUSI Final contains 660 lesion instances; BUS-UCLM Final contains 277 lesion instances from 35 scans.

– **BUS-UCLM:** BUS-UCLM provides 683 images from 38 scans acquired with a Siemens ACUSON S2000 system, including 419 normal, 174 benign, and 90 malignant images [19]. We mapped benign and malignant mask colors to a common lesion class, and restricted the external benchmark to lesion-positive images. Of the 264 lesion-positive candidates, four samples that failed quality-control checks were excluded. The final benchmark therefore contained 260 images from 35 scans and 277 lesion instances.

3.2 Implementation Settings and Evaluation

All datasets were standardized to the single-class YOLO polygon segmentation format. Multiple lesions within an image were retained as separate instances for Ultralytics validation, while their masks were merged by pixel-wise union for image-level evaluation. We rasterized ground-truth masks from the standardized polygons at the original image resolution. We used the same dataset-image keys across all models, resulting in 1,187 paired image-level observations per model.

We evaluated YOLOv8n-seg, YOLO11n-seg, and YOLO26n-seg [7-9]. To strictly assess the intrinsic learning capacity of each architecture without the confounding influence of external prior knowledge, all models were trained from scratch. Specifically, the networks were randomly initialized using their respective structural configurations (.yaml) rather than transferring pre-trained weights. Their fused checkpoints contained 3.258, 2.835, and 2.689 million parameters and required 11.3, 9.6, and 9.1 GFLOPs, respectively. For a fair comparison, all models were trained under the same configuration: up to 200 epochs, an image size of 640, a batch size of 32, early-stopping patience of 50, random seed 42, two data-loading workers, and GPU device 0. The optimizer was selected automatically by Ultralytics, with an initial learning rate of 0.01, final learning-rate factor of 0.01, momentum of 0.937, weight decay of 0.0005, and 3 warmup epochs. Data augmentation used mosaic (1.0), translation (0.1), scale (0.5), horizontal flipping (0.5), and RandAugment. Experiments were conducted on an NVIDIA Tesla P100 (16 GB) using Python 3.12, PyTorch 2.7.1+cu118, and Ultralytics 8.4.115. We

used BUS-BRA validation for training monitoring and selected the checkpoint with the highest validation mask mAP50:95: epoch 107 for YOLOv8n-seg, epoch 173 for YOLO11n-seg, and epoch 149 for YOLO26n-seg. Training ran for 157, 200, and 199 epochs, respectively, with early stopping patience of 50. The selected checkpoints were then frozen and evaluated on the BUS-BRA test, BUSI, and BUS-UCLM sets without further parameter updates.

We evaluated the model from two complementary perspectives. First, Ultralytics instance-level validation reported mask precision, mask recall, mAP@0.5, and mAP@0.5:0.95 [18]. Second, image-level segmentation was evaluated at `imgsz=640` with a fixed confidence threshold of 0.25 and `retina_masks=True`. We merged multiple predicted lesion masks within an image by pixel-wise union before comparing them with the corresponding ground-truth mask. Thus, mAP measures instance-level detection and segmentation across IoU thresholds, whereas Dice and IoU measure the overall lesion-region overlap [14].

For binary masks, let TP, FP, and FN denote the numbers of true-positive, false-positive, and false-negative pixels, respectively. Dice and IoU are defined as

$$\text{Dice} = \frac{2TP}{2TP + FP + FN}, \qquad \text{IoU} = \frac{TP}{TP + FP + FN}. \tag{1}$$

We computed both metrics per image and then averaged them within each dataset. We also calculated pixel-level precision and recall using standard binary definitions to support the analysis of segmentation failure patterns.

3.3 Statistical Analysis

For each model-dataset pair, 10,000 non-parametric image-level bootstrap resamples were drawn with replacement using seed 42 [2]. We recomputed mean Dice and IoU for each resample, and the 2.5th and 97.5th percentiles defined the 95% confidence intervals. We estimated uncertainty in the average generalization gap within the same bootstrap framework by recomputing the gap from the resampled mean Dice values.

Because the models were evaluated on identical image identifiers, Dice comparisons were paired. Two-sided Wilcoxon signed-rank tests compared YOLOv8 vs. YOLO11, YOLOv8 vs. YOLO26, and YOLO11 vs. YOLO26 within each dataset [23]. Holm correction controlled the family-wise error rate across the three comparisons, with adjusted $p < 0.05$ considered statistically significant [5]. The Dice-based generalization gap was defined from the source-domain mean Dice and the external mean Dice:

$$\begin{aligned} GG_D &= \overline{Dice}_{\text{BUS-BRA}} - \overline{Dice}_D, \\ \overline{Dice}_{ext} &= \frac{\overline{Dice}_{BUSI} + \overline{Dice}_{UCLM}}{2}, \\ GG_{avg} &= \overline{Dice}_{\text{BUS-BRA}} - \overline{Dice}_{ext}. \end{aligned} \tag{2}$$

Smaller gaps indicate better preservation of source-domain performance.

4 Experimental Results

4.1 Segmentation Performance

Table 2 shows the Ultralytics mask-level results. All three models remained strong on BUS-BRA, with mask mAP@0.5 above 0.97 and mAP@0.5:0.95 between 0.7342 and 0.7427. Performance was lower on both external datasets. YOLO11 achieved the highest mAP@0.5:0.95 on BUSI (0.4417), while YOLOv8 was highest on BUS-UCLM (0.4037). The change in ranking indicates that the model favored by an internal test set is not necessarily favored after transfer.

Table 2. Mask-level Segmentation Performance Across Datasets

Model	Dataset	Precision ↑	Recall ↑	mAP_{50} ↑	$mAP_{50:95}$ ↑
YOLOv8	BUS-BRA	0.9683	0.9539	0.9896	0.7409
	BUSI	0.7432	0.6924	0.7270	0.4257
	BUS-UCLM	0.8221	0.6674	0.7612	**0.4037**
YOLO11	BUS-BRA	0.9373	0.9681	0.9774	**0.7427**
	BUSI	0.8099	0.6879	0.7436	**0.4417**
	BUS-UCLM	0.8188	0.7076	0.7492	0.3775
YOLO26	BUS-BRA	0.9681	0.9645	0.9816	0.7342
	BUSI	0.8045	0.6546	0.7295	0.4274
	BUS-UCLM	0.7573	0.6968	0.7300	0.3551

Image-level Dice showed the same broad domain shift but a different model ranking (Table 3 and Fig. 3). BUS-BRA Dice ranged from 0.9010 to 0.9111, with YOLO26 numerically highest. On BUSI and BUS-UCLM, YOLO11 achieved the highest Dice, at 0.6520 and 0.6599, respectively. External Dice was considerably more variable, with standard deviations of approximately 0.32-0.38 on BUSI and BUS-UCLM, compared with 0.106–0.115 on BUS-BRA. This greater dispersion indicates that external performance was less consistent across images.

4.2 Generalization and Statistical Comparison

YOLO11 had the highest average external Dice (0.6560) and the smallest average generalization gap (0.2542), followed by YOLO26 (0.6251/0.2860) and YOLOv8 (0.5969/0.3041). The bootstrap 95% CIs for the average gap were 0.2266–0.2814 for YOLO11, 0.2561–0.3148 for YOLO26, and 0.2771–0.3312 for YOLOv8. The gap therefore leads to the same descriptive ordering as the mean external Dice while expressing the result relative to each model's source-domain baseline. The paired tests refine this descriptive ranking (Table 4). No comparison was significant on BUS-BRA. On BUSI, YOLO11 and YOLO26 each significantly

Table 3. Image-level Dice and IoU with Bootstrap 95% Confidence Intervals

Model	Dataset	Mean Dice (95% CI) ↑	Mean IoU (95% CI) ↑
YOLOv8n-seg	BUS-BRA	0.9010 (0.8865–0.9135)	0.8341 (0.8170–0.8495)
	BUSI	0.5888 (0.5609–0.6164)	0.5027 (0.4761–0.5288)
	BUS-UCLM	0.6049 (0.5660–0.6428)	0.5041 (0.4658–0.5415)
YOLO11n-seg	BUS-BRA	0.9102 (0.8967–0.9215)	0.8470 (0.8318–0.8611)
	BUSI	**0.6520 (0.6241–0.6793)**	**0.5709 (0.5448–0.5967)**
	BUS-UCLM	**0.6599 (0.6196–0.6998)**	**0.5689 (0.5305–0.6065)**
YOLO26n-seg	BUS-BRA	**0.9111 (0.8972–0.9226)**	**0.8489 (0.8336–0.8624)**
	BUSI	0.6332 (0.6045–0.6621)	0.5595 (0.5322–0.5869)
	BUS-UCLM	0.6170 (0.5718–0.6613)	0.5359 (0.4946–0.5779)

outperformed YOLOv8 after Holm correction, but YOLO11 and YOLO26 did not differ significantly. No BUS-UCLM comparison reached corrected significance. Thus, YOLO11 showed the most favorable external averages, but the evidence does not support a consistent statistical advantage over YOLO26.

Table 4. Paired Wilcoxon Signed-rank Tests with Holm Correction

Dataset	Comparison	Δ Mean Dice	p_{Holm}	Significant
BUS-BRA	YOLO11 vs. YOLOv8	+0.0092	0.1432	No
BUS-BRA	YOLO26 vs. YOLOv8	+0.0101	0.2472	No
BUS-BRA	YOLO11 vs. YOLO26	-0.0009	0.5905	No
BUSI	YOLO11 vs. YOLOv8	+0.0633	1.84×10^{-5}	Yes
BUSI	YOLO26 vs. YOLOv8	+0.0444	8.49×10^{-6}	Yes
BUSI	YOLO11 vs. YOLO26	+0.0188	0.7105	No
BUS-UCLM	YOLO11 vs. YOLOv8	+0.0550	0.5496	No
BUS-UCLM	YOLO26 vs. YOLOv8	+0.0120	0.7757	No
BUS-UCLM	YOLO11 vs. YOLO26	+0.0430	0.7757	No

5 Discussion

The main observation is not the small difference among models on BUS-BRA, but how their ranking changed after transfer. Source-domain Dice values were tightly clustered and none of the pairwise BUS-BRA comparisons was significant.

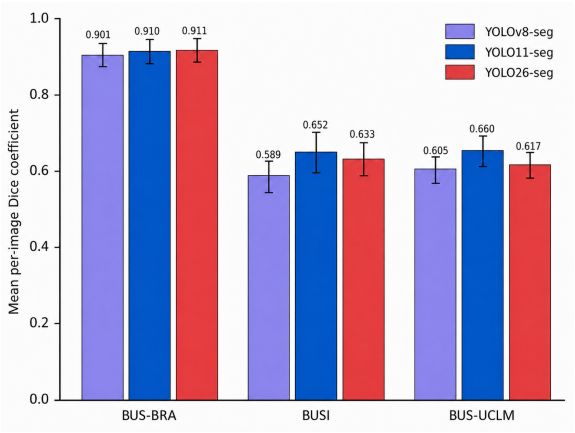


Fig. 3. Mean per-image Dice on BUS-BRA, BUSI, and BUS-UCLM. Error bars indicate bootstrap 95% confidence intervals.

YOLO26 therefore had a numerical, rather than statistically established, in-domain advantage. Once the same frozen checkpoints were evaluated externally, all three models degraded, and YOLO11 produced the strongest average Dice profile.

This pattern agrees with the cross-database degradation reported by Yu et al. on BUSI and BUS-UCLM [27], but the present setup isolates a different question by using BUS-BRA as a single development source. The external drop is also consistent with scanner-dependent variation reported in domain-generalization and test-time adaptation studies [10, 22]. Differences in probe characteristics, acquisition settings, operator technique, image preprocessing, and annotation style can all change the distribution seen by a segmentation model. Taken together, the results suggest that improvements measured on one BUS distribution do not necessarily survive changes in acquisition conditions.

YOLO26 illustrates this point particularly well. It combined the lowest computational complexity with the highest BUS-BRA Dice, yet YOLO11 achieved the highest Dice on both external datasets and the smallest average gap. The paired tests nevertheless prevent a stronger claim: YOLO11 did not significantly outperform YOLO26 after Holm correction on either external dataset. The appropriate interpretation is therefore that YOLO11 showed the most favorable descriptive generalization profile, not universal statistical superiority. This distinction matters when model differences are small in the source domain but become less stable after transfer.

The difference between mAP and Dice on BUS-UCLM is also informative. YOLOv8 achieved the highest mask mAP@0.5:0.95, whereas YOLO11 had the highest image-level Dice. These metrics answer different questions: Ultralytics mAP depends on instance matching across IoU thresholds, while the Dice analysis evaluates the union of predicted lesion pixels within each image [14]. A model can

therefore perform relatively well under one metric while being less favorable under another. Reporting both views provides a more complete picture of segmentation behavior.

Per-image analysis also revealed failure modes that dataset averages can hide. Complete misses produced Dice = 0 when no predicted lesion overlapped a non-empty ground truth. Over-segmentation reduced pixel precision when predicted regions extended beyond lesion boundaries, whereas under-segmentation reduced recall when only part of a lesion was captured. These patterns were more pronounced on the external datasets, where Dice variability was substantially greater than on BUS-BRA. This observation further supports interpreting complementary metrics rather than relying on a single scalar score [14].

Dataset curation should also be considered when interpreting the external results. BUSI normal images were excluded because the endpoint was lesion-positive segmentation, and one conflicting duplicate pair was removed after audit [13]. We combined multiple masks associated with one BUSI image so that all annotated lesion regions were retained. These choices improve consistency for lesion delineation, but they also change the evaluation question: the external benchmarks do not directly measure false-positive behavior in normal examinations. That distinction matters for clinical translation, where a detection-plus-segmentation system would need to be assessed on both lesion-positive and normal cases.

More broadly, the results reinforce the role of external validation in medical AI. A strong internal test result provides evidence that a model learned the development distribution, but it does not establish robustness to scanner or institutional variation [26]. For BUS segmentation, multi-center and scanner-diverse validation should therefore accompany claims of generalizability, especially when numerical gains on the source dataset are small.

6 Conclusion

This study compared three lightweight YOLO segmentation generations under a common-source cross-dataset protocol. All models performed strongly on BUS-BRA but degraded on BUSI and BUS-UCLM, demonstrating that in-domain performance does not necessarily predict external generalization. YOLO26 achieved the lowest model complexity and the highest in-domain Dice, whereas YOLO11 obtained the highest average external Dice and the smallest generalization gap. On BUSI, YOLO11 and YOLO26 significantly outperformed YOLOv8 after Holm correction, while the difference between YOLO11 and YOLO26 was not significant. These results highlight the importance of independent external datasets, uncertainty estimates, and paired statistical analysis when evaluating BUS lesion segmentation models. Several limitations should be considered. The study was restricted to three nano-scale YOLO variants with BUS-BRA as the single development source, and the external evaluation included only lesion-positive images. Each model was trained with a single random seed (42), so training-run variability and ranking stability were not assessed. Moreover, correlated frames in BUS-UCLM, polygon-derived ground-truth masks, and the fixed confidence threshold

may affect uncertainty estimates and model-specific performance. Future work should extend the common-source evaluation to alternative development datasets, larger YOLO scales, and multiple random seeds to assess source-dependence, scale effects, and ranking stability. Ultrasound-specific augmentation, scanner-diverse training, and feature-consistency learning should also be investigated to improve cross-dataset robustness without requiring target-domain labels.

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